

Polymer Simulation Knowledge Base

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Chapter 1

Polymer Simulation

A source-linked field guide to polymer informatics and molecular simulation, from representing polymer ensembles to building, parameterizing, simulating, and analyzing them.

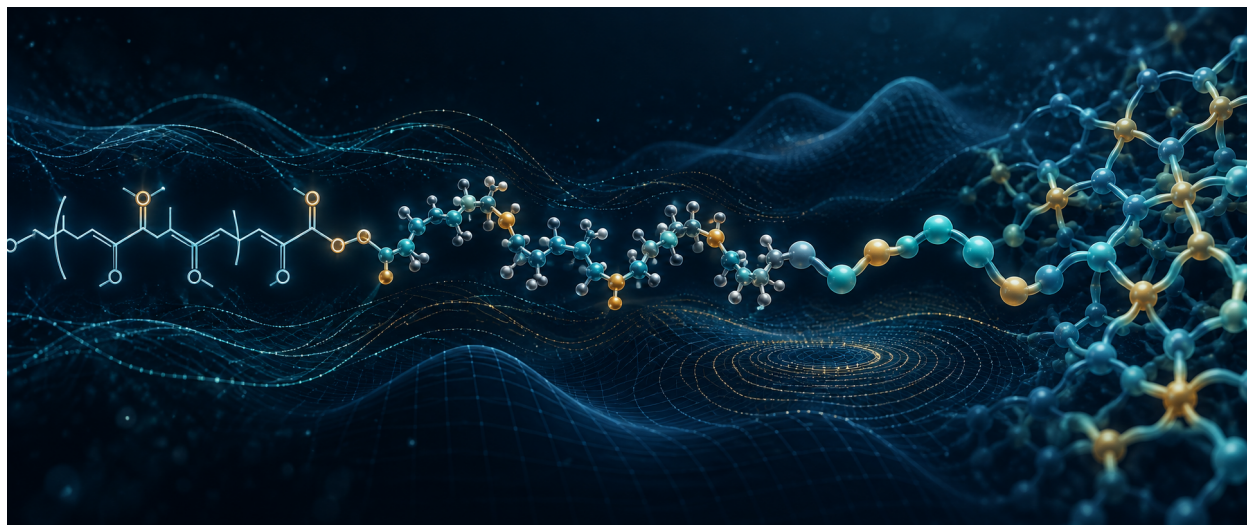


Figure 1.1: Polymer simulation cover: a polymer chain transitions from atomistic structure to coarse-grained model

1.1 Start with a task

If you need to...	Start here
Scan tools by workflow stage	Quick catalog below
Compare all 78 tools, sources, and evidence levels	Full tool catalog
Choose builders, force fields, MD engines, or analysis workflows	Practical tools guide
Read papers by topic and follow a suggested path	Literature guide

If you need to...	Start here
Read the guides as one document	PDF

1.2 Tool finder

Representative starting points from the 2026 catalog. Follow a category link for the complete list, descriptions, and sources.

Workflow stage	Quick picks	Full list
Polymer representation and data	BigSMILES · G-BigSMILES · CG-BigSMILES · PolyDAT · psmiles	Browse
Informatics, prediction, and generation	polyBERT · PolyMetriX · Polymer Genome · SMiPoly · PolyGraphMT · PolyGraphPy	Browse
Structure building and assembly	PSP · CHARMM-GUI · Polymer Builder · polyply · PolyConstruct · SwiftPol · HTPolyNet	Browse
Automated polymer simulation and design	RadonPy · PEMD · ADEPT · SPACIER · PolyRapid	Browse
ML force fields and agents	SimPoly / Vivace · PolyJarvis · CGMas	Browse
Force fields and topology tools	PolyParGen · Q-Force · foyer · GMSO · LigParGen · ParmEd	Browse
Coarse graining	VOTCA-CSG · PyCGTOOL · Swarm-CG · OpenMSCG · martinize2	Browse
Simulation engines	LAMMPS · GROMACS · OpenMM · HOOMD-blue · ESPResSo	Browse
Trajectory analysis and visualization	MDAnalysis · MDTraj · freud · OVITO	Browse
Workflow and data management	signac · signac-flow	Browse
Quantum and electronic structure	DFTB+ · Psi4 · xTB · CP2K	Browse

1.3 Scope

This repository covers polymer representations and datasets; structure construction; force-field assignment and coarse graining; atomistic, coarse-grained, and quantum calculations; workflow

automation; trajectory analysis; and polymer-focused machine learning and agent systems. The catalog distinguishes polymer-native projects from general software that supports polymer work.

Simulation results depend on chemistry, representation, parameterization, system preparation, and protocol. A software package or force-field family alone does not establish that a particular polymer model is validated.

1.4 Build

The PDF combines these Markdown guides. With Pandoc and XeLaTeX installed, run `make`. The same build runs through [GitHub Actions](#).

1.5 Contribute

Keep tool claims linked to primary project or publication sources. Label preprints and software-only entries clearly, and include version, parameter set, validation, and protocol details when documenting reproducible calculations. Add new Markdown chapters to `SOURCES` in the Makefile.

Chapter 2

Tools and Workflows

Use this page to choose a starting point. The [tool catalog](#) lists all 78 entries from the 2026 matrix by workflow stage; the [detailed reference](#) has practical notes on force fields, engines, selected workflows, and literature.

2.1 Follow the workflow

Stage	What to look for	Catalog section
Describe polymer chemistry and data	Stochastic representations, repeat-unit notation, identifiers, and polymer datasets	Representation, identifiers, and data
Predict or generate candidates	Polymer property models, embeddings, virtual libraries, and generative methods	Polymer informatics and generation
Build a molecular model	Polymer-aware chain, melt, electrolyte, branched, or network builders; general packing/assembly tools	Structure builders
Assign parameters and exchange topologies	Polymer-oriented parameterization plus general force-field and file-interoperability infrastructure	Parameterization and interoperability
Reduce resolution	Mapping and bottom-up or automated coarse-graining tools	Coarse graining
Run molecular simulation	General MD engines and specialized automated polymer workflows	Simulation engines · Automated workflows
Analyze structures and trajectories	Trajectory libraries, particle analysis, visualization, and workflow/data management	Analysis and visualization · Workflow management

Stage	What to look for	Catalog section
Use electronic-structure methods	DFT, tight-binding, and quantum-chemistry packages used for calculations or reference data	Quantum tools
Explore ML potentials and agents	Polymer MLFF work and emerging agent-assisted simulation/design systems	MLFFs and agents

2.2 Quick selection checks

1. **Start from the target material and property.** Identify whether the model is a homopolymer, copolymer, branched/cross-linked network, or electrolyte, and whether the target is structural, thermodynamic, transport, or mechanical.
2. **Separate the roles.** A builder creates a structure; a force field supplies interaction parameters; an engine integrates the equations of motion; a workflow may connect some of these steps. They are not interchangeable.
3. **Check representation and validation.** Confirm atomistic, united-atom, or coarse-grained resolution; chemistry and topology coverage; parameter provenance; and validation against relevant measurements or reference calculations.
4. **Check reproducibility before adopting a workflow.** Record the software version or commit, parameter set, input structure and mapping, dependencies, and simulation protocol. Review maintenance and licensing on the linked project page.

2.3 Scope and evidence

The catalog includes both polymer-native tools and general-purpose infrastructure that materially supports polymer modeling. Its **Core / Important / Supporting** labels are the supplied matrix’s navigation priorities, not independent rankings. **Evidence A** denotes a dedicated peer-reviewed paper plus a public project/code or documented tool; **B** denotes a documented project; **C** denotes general supporting infrastructure. These labels do not replace validation for a specific polymer chemistry.

For detailed force-field and model-representation notes, selected workflow comparisons, MLFF publications, and agent systems, continue to the [detailed reference](#). For foundational and topic-based reading, see the [papers guide](#).

Chapter 3

Polymer Informatics and Simulation Tool Catalog

This catalog maps the tools in the supplied 2026 literature matrix to the stage of a polymer modeling workflow. It includes polymer-specific packages and general-purpose infrastructure used to construct, parameterize, simulate, or analyze polymer systems.

Scope and evidence fields are transcribed from the supplied matrix (dated 2026-10-07). Links and publication metadata should be checked against the linked project or publisher before citation or production use. A publication or project listing is not, by itself, evidence that a parameter set is valid for a target polymer.

3.1 How to read the catalog

- **Core / Important / Supporting** reproduces the matrix priority labels; it is a curation aid, not a quality ranking.
- **Evidence A** means a dedicated peer-reviewed paper plus a public project/code or clearly documented tool; **B** means a documented software/project; **C** means supporting/general infrastructure.
- Capability depends on each tool's input, model representation, chemistry coverage, dependencies, and release. Check the project documentation before choosing a workflow.

3.2 Representation, identifiers, and data

Tool	Role in a workflow	Evidence / priority	Publication or project source
BigSMILES	Machine-readable representation of stochastic macromolecular ensembles	A / Core	BigSMILES: A Structurally-Based Line Notation for Describing Macromolecules (2019) · code · docs

Tool	Role in a workflow	Evidence / priority	Publication or project source
bigsmiles (Python parser)	Parse, validate and manipulate BigSMILES strings	B / Important	project / software source
G-BigSMILES	Augment BigSMILES with ensemble-generation information such as connection probabilities and molecular-weight distributions	A / Core	Generative BigSMILES: an extension for polymer informatics, computer simulations & ML/AI (2024)
CG-BigSMILES	Line notation for coarse-grained polymer models including mapping and force-field links	A / Core	CG-BigSMILES: Line Notation for Coarse-Grained Models of Polymers (2025)
PolyDAT	Standard schema for polymer characterization, synthesis and transformations	A / Core	PolyDAT: A Generic Data Schema for Polymer Characterization (2021)
psmiles	Canonicalize, randomize, dimerize, fingerprint and form alternating copolymers	B / Core	project / software source
canonicalize_psmiles	Canonicalize repeat-unit polymer SMILES	B / Important	project / software source

3.3 Polymer informatics, property prediction, and generation

Tool	Role in a workflow	Evidence / priority	Publication or project source
polyBERT	Generate polymer embeddings/fingerprints and enable rapid property prediction	A / Core	polyBERT: a chemical language model to enable fully machine-driven ultrafast polymer informatics (2023) · code · docs

Tool	Role in a workflow	Evidence / priority	Publication or project source
PolyMetriX	Data acquisition, polymer featurization, splitting, modeling and evaluation	A / Core	PolyMetriX: an ecosystem for digital polymer chemistry (2025)
Polymer Genome	Web-based ML predictions for polymer properties	A / Core	Polymer Genome: A Data-Powered Polymer Informatics Platform for Property Predictions (2018)
SMiPoly	Generate synthesizable virtual polymer libraries using polymerization reaction rules	A / Core	SMiPoly: Generation of a Synthesizable Polymer Virtual Library Using Rule-Based Polymerization Reactions (2023)
copolymer_informatics	Train multi-task neural networks for copolymer property prediction	A / Important	project / software source; Copolymer Informatics with Multi-Task Deep Neural Networks (2022)
PolymerAI	Random polymer generation and reinforcement-learning inverse design	B / Important	project / software source; PolymerAI software release (2020)
PolyGraphMT	Joint prediction of multiple polymer properties across fidelity levels	A / Core	ADEPT-PolyGraphMT: automated molecular simulation and multi-task multi-fidelity machine learning for polymer property generation and prediction (2026)
PolyGraphPy	Automate DFTB data generation, Bayesian GNN prediction and property-guided generation	A / Core	PolyGraphPy: A unified Python framework for atomistic simulation and machine learning-driven polymer design (2026)

Tool	Role in a workflow	Evidence / priority	Publication or project source
TRI-AMDD PolyGen	GPT/diffusion-based generation of polymer electrolytes and iterative discovery	B / Important	De novo design of polymer electrolytes with high conductivity using generative AIs (2023)

3.4 Polymer structure builders and molecular assembly

Tool	Role in a workflow	Evidence / priority	Publication or project source
polyGen (3D)	Generate atomic-level polymer conformations from repeat-unit chemistry	A / Core	A Learning Framework for Atomic-Level Polymer Structure Generation (2025)
PSP (Polymer Structure Predictor)	Build oligomers, infinite chains, crystals and amorphous polymer models	A / Core	Polymer Structure Predictor (PSP): A Python Toolkit for Predicting Atomic-Level Structural Models for a Range of Polymer Geometries (2022)
CHARMM-GUI Polymer Builder	Build relaxed polymer chains, melts, solutions and block-copolymer systems	A / Core	CHARMM-GUI Polymer Builder for Modeling and Simulation of Synthetic Polymers (2021)
pysimm	Build, modify, parameterize and simulate molecular systems; strong amorphous-polymer support	A / Core	pysimm: A python package for simulation of molecular systems (2017)
Polymatic	Build polymeric networks by iterative bond formation with relaxation	A / Core	Polymatic: a generalized simulated polymerization algorithm for amorphous polymers (2013)

Tool	Role in a workflow	Evidence / priority	Publication or project source
polyply	Generate parameters and coordinates for complex macromolecular topologies at arbitrary resolution	A / Core	Polyply; a python suite for facilitating simulations of macromolecules and nanomaterials (2022)
pyPolyBuilder	Generate topologies and initial structures for linear and branched supramolecules	A / Important	pyPolyBuilder: Automated Preparation of Molecular Topologies and Initial Configurations for Molecular Dynamics Simulations of Arbitrary Supramolecules (2021)
PolyConstruct	Generate polymer coordinates and topology files by adapting biomolecular simulation pipelines	A / Core	PolyConstruct: Adapting Biomolecular Simulation Pipelines for Polymers with PolyBuild, PolyConf, and PolyTop (2025)
SwiftPol	Build and parameterize polymer systems while explicitly controlling molecular-weight distribution and sequence features	A / Core	SwiftPol: A Python package for building and parameterizing in silico polymer systems (2025)
PySoftK	Build polymer topologies and analyze interfaces, aggregation, interactions and self-assembly	A / Core	Automated Analysis of Soft Matter Interfaces, Interactions, and Self-Assembly with PySoftK (2025) · code · docs
MolPy	Modular construction, sequence handling, topology editing, force-field assignment, packing and multi-engine export	A / Core	MolPy: A Large Language Model-Friendly Toolkit for Reactive Topology Editing in Polymer Simulations (2026)

Tool	Role in a workflow	Evidence / priority	Publication or project source
PolymerModeler	Create amorphous polymer systems and run MD/property calculations	B / Important	PolymerModeler: A general-purpose simulation tool for polymers (2015)
mBuild	Build complex molecular systems from reusable components	A / Important	A Hierarchical, Component Based Approach to Screening Properties of Soft Matter (2016) · code · docs
Moltemplate	Prepare complex atomistic or coarse-grained molecular systems for LAMMPS	A / Important	Moltemplate: A Tool for Coarse-Grained Modeling of Complex Biological Matter and Soft Condensed Matter Physics (2021) · code · docs
Packmol	Pack molecules into simulation boxes subject to geometric constraints	A / Supporting	PACKMOL: A package for building initial configurations for molecular dynamics simulations (2009)
Assemble!	Create polymer chains from monomers and assemble solvated/mixture systems for GROMACS	A / Important	Easy creation of polymeric systems for molecular dynamics with Assemble! (2016)
EMC (Enhanced Monte Carlo)	Build and equilibrate atomistic polymer systems using Monte Carlo/connectivity methods	B / Important	project / software source ; Enhanced Monte Carlo polymer building software
XPB	High-throughput generation of linear, dendritic, hyperbranched and crosslinked polymer structures	A / Core	XPB: an Extendable Polymer Builder for High-Throughput and High-Quality Generation of Complex Polymer Structures (2025)

Tool	Role in a workflow	Evidence / priority	Publication or project source
HTPolyNet	Generate atomistic crosslinked polymer networks and GROMACS topology files	A / Core	HTPolyNet: A general system generator for atomistic simulations of polymer networks (2023)
Chameleon	Efficient phase-space sampling of dense realistic polymers using connectivity-altering MC moves	A / Important	Chameleon: A generalized, connectivity altering software for tackling properties of realistic polymer systems (2019)

3.5 Automated simulation and design workflows

Tool	Role in a workflow	Evidence / priority	Publication or project source
PEMD	End-to-end modeling of solid polymer electrolytes with OPLS-AA, QM, GROMACS and transport analysis	A / Core	PEMD: An open-source framework for high-throughput simulation and analysis of polymer electrolytes (2026)
RadonPy	Automate modeling, charge calculation, FF assignment, equilibration, NEMD and property extraction	A / Core	RadonPy: automated physical property calculation using all-atom classical molecular dynamics simulations for polymer informatics (2022)
ADEPT	SMILES-to-properties high-throughput workflow combining amorphous model generation, LAMMPS and Psi4	A / Core	ADEPT-PolyGraphMT: automated molecular simulation and multi-task multi-fidelity machine learning for polymer property generation and prediction (2026)

Tool	Role in a workflow	Evidence / priority	Publication or project source
SPACIER	Closed-loop/on-demand polymer design using RadonPy simulation inside Bayesian optimization	A / Core	SPACIER: on-demand polymer design with fully automated all-atom classical molecular dynamics integrated into machine learning pipelines (2025)
PolyRapid	Automated polymer melt equilibration and high-throughput screening with convergence criteria	A / Core	PolyRapid: automated high-throughput screening of polymers using a computational workflow (2026)

3.6 Machine-learned force fields and agentic systems

Tool	Role in a workflow	Evidence / priority	Publication or project source
SimPoly / Vivace	Run polymer MD with first-principles-derived MLFF; includes PolyPack/PolyDiss data and Tg/density protocols	A / Core	SimPoly: Simulation of Polymers with Machine Learning Force Fields Derived from First Principles (2025) · code · docs
PolyJarvis	Natural-language-to-polymer-MD agent using MCP servers and RadonPy/EMC/LAMMPS	A / Core	PolyJarvis: LLM Agent for Autonomous Polymer MD Simulations (2026)
CGMas	Automate AA topology, equilibration, AA→CG mapping, potential derivation and validation from natural language	A / Core	A Multi-Agent Framework for Automated Coarse-Grained Molecular Dynamics of Polymers (2026)

3.7 Force-field parameterization and topology interoperability

Tool	Role in a workflow	Evidence / priority	Publication or project source
PolyPal	Assist parameterization and construction of amorphous all-atom polymer simulations	B / Important	project / software source
PolyParGen	Generate MD force-field parameters for polymer repeat structures and convert oligomer parameters to polymer form	A / Core	Development of PolyParGen Software to Facilitate the Determination of Molecular Dynamics Simulation Parameters for Polymers (2019)
Q-Force	Derive bonded parameters and charges from QM while retaining transferable nonbonded terms	A / Core	Q-Force: Quantum Mechanically Augmented Molecular Force Fields (2021)
foyer	Define, disseminate and apply force-field atom-typing rules reproducibly	A / Important	Formalizing atom-typing and the dissemination of force fields with foyer (2019) · code · docs
GMSO	Engine-agnostic storage and conversion of molecular topology and force-field information	B / Important	project / software source ; GMSO software / MoSDeF ecosystem · code · docs
OpenFF Toolkit	Assign modern small-molecule force fields and manipulate chemical topologies; polymer support evolving	B / Important	project / software source ; Open Force Field Initiative software · code · docs
OpenFF Interchange	Convert parameterized systems between OpenFF and simulation engines	B / Supporting	project / software source ; OpenFF Interchange software · code · docs
ForceBalance	Optimize classical force-field parameters against QM/experimental reference data	A / Important	Systematic Parametrization of Polarizable Force Fields from Quantum Chemistry Data (2013) · code · docs

Tool	Role in a workflow	Evidence / priority	Publication or project source
AmberTools / Antechamber	Assign GAFF/GAFF2 atom types and derive small-molecule parameters used in many polymer workflows	B / Supporting	project / software source ; AmberTools / Antechamber canonical software
LigParGen	Generate OPLS-AA parameters for organic molecules; often used on monomers/oligomers	A / Supporting	LigParGen web server literature (2017)
CGenFF / ParamChem	Assign CGenFF atom types and parameters to organic molecules	A / Supporting	CHARMM General Force Field literature (2010)
ParmEd	Read, edit and convert molecular mechanics topology/parameter formats	B / Supporting	project / software source ; ParmEd software · code · docs
InterMol	Convert molecular simulation inputs among major MD engines	A / Supporting	InterMol: A program for molecular simulation file conversions (2014)

3.8 Coarse-graining tools

Tool	Role in a workflow	Evidence / priority	Publication or project source
VOTCA-CSG	Derive coarse-grained potentials using IBI, force matching and related methods	A / Important	Versatile Object-oriented Toolkit for Coarse-graining Applications (VOTCA) (2009) · code · docs
PyCGTOOL	Generate CG bonded parameters from atomistic trajectories using Boltzmann inversion	A / Important	PyCGTOOL: Automated Generation of Coarse-Grained Molecular Dynamics Models from Atomistic Trajectories (2017) · code · docs

Tool	Role in a workflow	Evidence / priority	Publication or project source
Swarm-CG	Optimize Martini bonded terms against AA reference trajectories using particle swarm optimization	A / Important	Swarm-CG: Automatic Parametrization of Bonded Terms in MARTINI-based Coarse-Grained Models (2020)
OpenMSCG	Build CG models via force matching, relative entropy and related methods	B / Supporting	project / software source ; OpenMSCG software · code · docs
vermouth	Graph transformations and topology processing underlying martinize2/polyply	B / Supporting	project / software source ; Vermouth software / Martini ecosystem
martinize2	Generate Martini coarse-grained topologies from atomistic structures	B / Supporting	project / software source ; martinize2 software · code · docs

3.9 Simulation engines

Tool	Role in a workflow	Evidence / priority	Publication or project source
HOOMD-blue	High-performance GPU MD and particle simulation used extensively in soft matter and coarse-grained polymers	A / Supporting	HOOMD-blue software papers (2020) · code · docs
ESPResSo	Particle-based soft-matter simulation with electrostatics, hydrodynamics and polymer models	A / Supporting	ESPResSo software papers (2019) · code · docs

Tool	Role in a workflow	Evidence / priority	Publication or project source
LAMMPS	General MD engine; dominant backend in automated polymer workflows such as RadonPy/ADEPT/SimPoly	A / Supporting	LAMMPS - a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales (2022) · code · docs
GROMACS	High-performance MD engine widely used for atomistic/CG polymer simulations	A / Supporting	GROMACS development and performance papers (2020) · code · docs
OpenMM	Python-friendly GPU molecular simulation engine useful for automated workflows	A / Supporting	OpenMM 7: Rapid development of high performance algorithms for molecular dynamics (2017) · code · docs

3.10 Trajectory analysis and visualization

Tool	Role in a workflow	Evidence / priority	Publication or project source
MDAnalysis	Read, transform and analyze MD trajectories across engines	A / Supporting	MDAnalysis: A Python package for the rapid analysis of molecular dynamics simulations (2016) · code · docs
MDTraj	Fast trajectory I/O and common structural analyses	A / Supporting	MDTraj: A Modern Open Library for the Analysis of Molecular Dynamics Trajectories (2015) · code · docs
freud	High-performance RDF, clustering, order parameters and local-environment analysis	A / Supporting	freud: A software suite for high throughput analysis of particle simulation data (2020) · code · docs
OVITO	Interactive and scripted structural analysis/visualization of particle simulations	A / Supporting	OVITO - a scientific data visualization and analysis software (2010)

3.11 Workflow and data management

Tool	Role in a workflow	Evidence / priority	Publication or project source
signac	Manage parameter spaces, jobs and simulation data in high-throughput computational studies	A / Supporting	signac: A Python framework for data and workflow management (2018) · code · docs
signac-flow	Define and submit reproducible simulation workflows on local/HPC schedulers	B / Supporting	project / software source; signac-flow software · code · docs

3.12 Quantum and electronic-structure tools

Tool	Role in a workflow	Evidence / priority	Publication or project source
DFTB+	Fast approximate quantum calculations used for polymer/monomer datasets and electronic properties	A / Supporting	DFTB+, a software package for efficient approximate density functional theory based atomistic simulations (2020) · code · docs
Psi4	Electronic structure, charge, polarizability and other QM calculations	A / Supporting	Psi4 1.4: Open-source software for high-throughput quantum chemistry (2020) · code · docs
xTB	Fast geometry optimization, conformer refinement and electronic estimates	A / Supporting	Extended tight-binding quantum chemistry methods/software (2017) · code · docs
CP2K	Periodic DFT, ab initio MD and mixed quantum/classical simulations for condensed polymer systems	A / Supporting	CP2K: An electronic structure and molecular dynamics software package (2020) · code · docs

3.13 Related reference

For force-field selection notes, modeling caveats, detailed descriptions, and selected polymer simulation and MLFF papers, see the [practical tools guide](#). The separate [literature guide](#) provides foundational reading paths.

Chapter 4

Tools and References

This guide separates force-field models, simulation engines, polymer-building workflows, and research papers. A package may automate a workflow, but it does not make the underlying model valid for every polymer: check each force field’s chemical coverage and compare calculated properties with experiment or a trusted reference.

4.1 Classical Force Fields and Model Representations

Name	What it describes	Notes for polymer work
AMBER	A family of biomolecular force fields and simulation programs. The related GAFF extends AMBER-style parameters to common organic molecules.	GAFF: Wang, Wolf, Caldwell, Kollman & Case, “Development and testing of a general amber force field,” <i>Journal of Computational Chemistry</i> 25 (9), 1157–1174 (2004), DOI . Do not treat protein/nucleic-acid parameter sets as general polymer parameters; see AMBER documentation .

Name	What it describes	Notes for polymer work
CHARMM	A force-field family and simulation program, with parameter sets for biomolecules, lipids, and selected small molecules.	CHARMM can mean either the force-field family or the simulation engine. CGenFF: Vanommeslaeghe, Hatcher, Acharya et al., "CHARMM general force field: A force field for drug-like molecules compatible with the CHARMM all-atom additive biological force fields," <i>Journal of Computational Chemistry</i> 31 (4), 671–690 (2010), DOI. Check its chemical coverage before polymer use.
OPLS-AA	An all-atom potential developed for conformational energetics and organic-liquid properties.	Jorgensen, Maxwell & Tirado-Rives, "Development and testing of the OPLS all-atom force field on conformational energetics and properties of organic liquids," <i>Journal of the American Chemical Society</i> 118 (45), 11225–11236 (1996), DOI. Validate torsions, charges, and condensed-phase properties for the polymer of interest.
PCFF	A class-II, all-atom force-field family developed for synthetic polymers and related materials. The linked study applies PCFF to poly(epsilon-caprolactone); the CFF93 polycarbonate paper is a related foundational source.	Confirm that the installed parameter set covers the chemistry and properties being studied. PCFF and COMPASS are related CFF-family models, but their parameters are not interchangeable.

Name	What it describes	Notes for polymer work
GROMOS	A biomolecular force-field family and simulation package; its parameter sets are tuned for particular molecular classes and condensed-phase data.	Representative force-field reference: Oostenbrink, Villa, Mark & van Gunsteren, "A biomolecular force field based on the free enthalpy of hydration and solvation: the GROMOS force-field parameter sets 53A5 and 53A6," <i>Journal of Computational Chemistry</i> 25 (13), 1656–1676 (2004), DOI . Not a universal synthetic-polymer force field.
COMPASS	An all-atom, class-II force field parameterized for condensed-phase properties of organic, inorganic, and polymeric materials.	Sun, "COMPASS: An ab initio force-field optimized for condensed-phase applications—overview with details on alkane and benzene compounds," <i>The Journal of Physical Chemistry B</i> 102 (38), 7338–7364 (1998), DOI . Check installed version and parameter coverage.
ReaxFF	A bond-order reactive force field that can represent bond formation and breaking.	Foundational paper: van Duin, Dasgupta, Lorant & Goddard, "ReaxFF: A Reactive Force Field for Hydrocarbons," <i>The Journal of Physical Chemistry A</i> 105 (41), 9396–9409 (2001), DOI . Validate the parameter set for the relevant elements, bonding, and reaction conditions.
United-atom (UA)	A representation that combines many carbon-bound hydrogens with the carbon into one interaction site; TraPPE-UA is one named parameter family.	UA is a level of representation, not a force field by itself. Polar hydrogens and heteroatoms may remain explicit, depending on the model.

Name	What it describes	Notes for polymer work
Coarse-grained (CG)	A representation that groups atoms into larger interaction sites. Kremer & Grest, “Dynamics of entangled linear polymer melts: A molecular-dynamics simulation,” <i>The Journal of Chemical Physics</i> 92 , 5057–5086 (1990), introduced a generic bead-spring polymer-melt model; MARTINI 3 is a chemistry-informed CG force field.	CG models trade chemical detail for larger length and time scales. State the mapping and target properties; a generic model is not automatically transferable to a specific polymer.

4.2 Molecular-Dynamics and Modeling Software

Software	Main use	Links
LAMMPS	Open-source, parallel classical MD for polymers, atomistic materials, and coarse-grained systems. Supports many force-field styles, including class-II and reactive models.	Thompson et al., “LAMMPS—A flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales,” <i>Computer Physics Communications</i> 271 , 108171 (2022), DOI · documentation
GROMACS	Open-source, high-performance MD and trajectory analysis; commonly used for biomolecules and also used for polymer systems.	Abraham et al., “GROMACS: High performance molecular simulations through multi-level parallelism from laptops to supercomputers,” <i>SoftwareX</i> 1–2 , 19–25 (2015), DOI · documentation · tutorials
NAMD	Parallel MD software designed for large biomolecular systems.	Phillips et al., “Scalable molecular dynamics on CPU and GPU architectures with NAMD,” <i>The Journal of Chemical Physics</i> 153 , 044130 (2020), DOI · user guide

Software	Main use	Links
CHARMM	Molecular simulation and analysis program that implements CHARMM force fields and other modeling methods.	Current documentation
BIOVIA Materials Studio	Commercial materials-modeling environment for building structures and running atomistic, mesoscale, and property calculations.	Product modules and capabilities depend on the licensed installation.

4.3 Polymer Construction, Simulation, and Analysis Workflows

Tool	What it helps with	Source
RadonPy	Open-source Python workflow that automates all-atom classical MD property calculations, including model construction, force-field assignment, equilibration checks, and post-processing. Check the repository for the current release and supported dependencies.	Hayashi, Shiomi, Morikawa & Yoshida, “RadonPy: automated physical property calculation using all-atom classical molecular dynamics simulations for polymer informatics,” <i>npj Computational Materials</i> 8 , 222 (2022), DOI · Documentation
pysimm	Open-source Python toolkit for building/modifying molecular systems, assigning force fields, and connecting workflows to LAMMPS and other engines; includes a polymer random-walk builder.	Fortunato & Colina, “pysimm: A Python Package for Simulation of Molecular Systems,” <i>SoftwareX</i> 6 , 7–12 (2017), DOI · current code/version: project documentation · v1.1.1 archive (2025)
PySoftK	Open-source Python package for building polymer chains and analyzing soft-matter simulations, including interfaces, aggregation/self-assembly, and intermolecular interactions. The cited v1.0 paper is primarily an analysis toolkit paper.	López-Ríos de Castro, Santana-Bonilla, Ziólek & Lorenz, “Automated Analysis of Soft Matter Interfaces, Interactions, and Self-Assembly with PySoftK,” <i>Journal of Chemical Information and Modeling</i> 65 (4), 1679–1684 (2025), DOI · Documentation

Tool	What it helps with	Source
SMiPoly	Rule-based generation of virtual polymer structures from monomers and polymerization reactions; the published rules target linear polymers, not arbitrary branched architectures.	Ohno, Hayashi, Zhang, Kaneko & Yoshida, "SMiPoly: Generation of a Synthesizable Polymer Virtual Library Using Rule-Based Polymerization Reactions," <i>Journal of Chemical Information and Modeling</i> 63 (17), 5539–5548 (2023), DOI · Documentation
SPACIER	Open-source design workflow integrating RadonPy all-atom property calculations and Bayesian optimization; demonstrated for optical polymer design.	Nanjo, Arifin, Maeda, Hayashi, Hatakeyama-Sato, Himeno, Hayakawa & Yoshida, "SPACIER: on-demand polymer design with fully automated all-atom classical molecular dynamics integrated into machine learning pipelines," <i>npj Computational Materials</i> 11 , 16 (2025), DOI · code · arXiv version
PEMD	Open-source Python framework unifying polymer construction, force-field parameterization, multiscale simulation, and electrolyte property analysis. The paper reports 656 successfully constructed homopolymers and screens 200 electrolyte systems.	Tan, Liang, Lu, Ji, Ji, Li & Hou, "PEMD: a high-throughput simulation and analysis framework for solid polymer electrolytes," <i>Digital Discovery</i> 5 (1), 193–202 (2026), DOI · code · archived release
MolPy	Open-source toolkit for inspectable, modular molecular-topology construction and editing, including reactive/cross-linking polymer workflows and structured interfaces for agent-assisted coding.	Li, Schwarz, Guo, Sun & Brandell, "MolPy: A Large Language Model-Friendly Toolkit for Reactive Topology Editing in Polymer Simulations," <i>Journal of Chemical Information and Modeling</i> 66 (13), 7562–7572 (2026), DOI

Tool	What it helps with	Source
PolyRapid	Adaptive automated annealing and screening demonstrated on 103 homopolymers; reports 95% equilibration within four 30 ns cycles and a public dataset for downstream modeling.	Smith, Ericson, Fantauzzo, Yong, Carbone & Troisi, “PolyRapid: automated high-throughput screening of polymers using a computational workflow,” <i>Soft Matter</i> 22 (33), 5480–5492 (2026), DOI · open full text
ADEPT / PolyGraphMT	ADEPT builds atomistic polymer models and computes properties; PolyGraphMT learns across property tasks and data fidelities using simulation, experiment, and group-contribution data.	Alosious, Liu, Xu, Liu, Zhang, Jiang & Luo, “ADEPT-PolyGraphMT: automated molecular simulation and multi-task multi-fidelity machine learning for polymer property generation and prediction,” <i>Digital Discovery</i> 5 (7), 3021–3043 (2026), DOI · preprint · code
PolyGraphPy	Open-source Python framework using DFTB+ data generation, Bayesian graph neural networks with uncertainty, and SELFIES/BRICS generative models for polymer property prediction and design.	Duarte, Venkatram, Cencer, Dumitrică & dos Santos, “PolyGraphPy: A unified Python framework for atomistic simulation and machine learning-driven polymer design,” arXiv:2606.06415v2 (2026; preprint as of 7 Oct 2026), arXiv · related journal article scheduled for 25 Oct 2026: <i>Computational Materials Science</i> 275 , 114985 · code

4.4 Force-Field Parameterization

Tool	Main use	Source
Sobtop	Desktop utility for supported small molecules that prepares force-field parameters and GROMACS topology files; the project page should be consulted for supported input formats and current release.	Official project/download page · cite software with the version/date actually used; no journal article located.

Tool	Main use	Source
LigParGen	Web server that assigns OPLS-AA parameters to supported organic ligands from a molecular structure/SMILES. It does not automatically validate polymer transferability.	Dodda, Cabeza de Vaca, Tirado-Rives & Jorgensen, “LigParGen web server: an automatic OPLS-AA parameter generator for organic ligands,” <i>Nucleic Acids Research</i> 45 (W1), W331–W336 (2017), DOI · Yale service
PolyParGen	Open-source workflow that partitions a polymer into supported fragments, obtains their parameters, and combines them into a polymer GROMACS topology; tested on P3EHT and F-P3EHT in chloroform.	Yabe, Mori, Ueda & Takeda, “Development of PolyParGen Software to Facilitate the Determination of Molecular Dynamics Simulation Parameters for Polymers,” <i>Journal of Computer Chemistry, Japan—International Edition</i> 5 , 2018-0034 (2019), DOI · article

4.5 Machine-Learned Force Fields (MLFFs) and Polymer-Property Papers

Paper	What it contributes	Publication status
SimPoly: Simulation of Polymers with Machine Learning Force Fields Derived from First Principles	Introduces Vivace, an ML force field trained from first-principles data, and PolyArena, an experimental bulk-property benchmark spanning 130 polymers; reports density and glass-transition predictions.	Simm, Hélie, Schulz, Chen, Simeon, Kuzina, Martinez-Baez, Gasparotto, Tocci, Chen, Li, Cheng, Wang, Nguyen, Smith & Sun, arXiv:2510.13696 (v1 submitted 15 Oct 2025; preprint), arXiv DOI · code and model
On-the-Fly Machine-Learned Force Fields for High-Fidelity Polymer Glass Transition Simulations	Adaptively calls first-principles calculations when configurations leave the model’s confidence domain; demonstrates near-quantum Tg simulations on 12 polymers and reports roughly six orders lower cost than AIMD.	Srivastava, Agarwal, Shukla, Sahu & Ramprasad, <i>Journal of Chemical Information and Modeling</i> 66 (17), 11129–11139 (2026), DOI · arXiv:2601.17137

Paper	What it contributes	Publication status
Active Learning of a Neural Network Potential for Large-Scale Atomistic Simulations of Polymer Electrolyte Membranes	Active-learning deep potential for Nafion over hydration states; enables 10,000–20,000-atom simulations to 31 ns and tests transfer to hydration level $\lambda = 24$ beyond the training range.	Yoshimoto, Matsumura, Yamazaki, Iwasaki, Nakao & Sakai, <i>The Journal of Physical Chemistry B</i> 130 (36), 9326–9337 (2026), DOI
How Long Is Long Enough? Extrapolation of Machine-Learning Interatomic Potentials for Oligomeric and Polymeric Systems	Controlled n-alkane training study (n=1–8): transfer improves when local chemical environments converge; careful neighbor-list construction improves intermolecular-energy learning.	Hooven, Lin, Carroll & Cersonsky, <i>Journal of Chemical Theory and Computation</i> 22 (13), 6917–6926 (2026), DOI
From Oligomers to Entangled Polymers: How to Train a Transferable Machine Learning Interatomic Potential	Compares descriptors and active learning for polyethylene; reports ACE transfer from small oligomers and evaluates non-bonded interactions against OPLS-AA trajectories.	Fischer & Heuer, arXiv:2608.01162 (submitted 2 Aug 2026; preprint), arXiv
Glass Transition Prediction of Binary Copolymers Across Large Chemical Spaces Using Machine Learning and Physics-Based Modeling	Composition-aware models trained on experimental data report RMSE about 14 K and R-squared about 0.98; screens about 148,000 binary copolymer compositions and validates with MD.	Bhati, Afzal, Chew, Browning & Halls, <i>Polymers</i> 18 (14), 1727 (2026), DOI
Rapid, Accurate and Reproducible Prediction of the Glass Transition Temperature Using Ensemble-Based Molecular Dynamics Simulation	Quantifies replica uncertainty for six cross-linked epoxy resins; concurrent-temperature simulation agrees with sequential cooling, and 10 or more replicas support 95% confidence intervals below 20 K.	Suter, Müller, Vassaux, Anastasiou, Simmons, Tilbrook & Coveney, <i>Journal of Chemical Theory and Computation</i> 21 (3), 1405–1421 (2025), DOI
PolyRapid: Automated High-Throughput Screening of Polymers Using a Computational Workflow	Screens 103 homopolymers with adaptive annealing; reports 95% equilibrated within four 30 ns cycles, 100% within nine, and ML classification for density/T _g .	Smith, Ericson, Fantauzzo, Yong, Carbone & Troisi, <i>Soft Matter</i> 22 (33), 5480–5492 (2026), DOI

Paper	What it contributes	Publication status
Dissipative Particle Dynamics Modeling in Polymer Science and Engineering	Review of DPD methods, mapping and polymer applications spanning blends, solutions, self-assembly, interfaces, membranes, rheology, and composites.	Nikkhah & Vandichel, <i>WIREs Computational Molecular Science</i> 15 (2), e70018 (2025), DOI
Atomistic Polymer Modeling: Recent Advances and Challenges in Building and Parametrization Workflows	Reviews synthetic-polymer system-building/parameterization gaps, FAIR practice, cheminformatics, direct chemical perception, and neural-network workflows.	Turney & Matta, <i>Macromolecules</i> 58 (21), 11509–11522 (2025), DOI
ADEPT-PolyGraphMT: Automated Molecular Simulation and Multi-Task Multi-Fidelity Machine Learning for Polymer Property Generation and Prediction	Integrates atomistic simulation with curated experimental and group-contribution values into ~62,000 records for 28 properties; benchmarks multi-task and multi-fidelity learning.	Alosious, Liu, Xu, Liu, Zhang, Jiang & Luo, <i>Digital Discovery</i> 5 (2026), DOI · preprint

4.6 Polymer Discovery Agents and Representations

Resource	Scope	Source / status
Polymer-Agent	LLM-driven terminal workflow for polymer property prediction, property-guided structure generation, and candidate modification; a research preprint, not a validated general-purpose autonomous lab.	Nigam, Chandrasekhar & Farimani, “PolyAgent: Large Language Model Agent for Polymer Design,” arXiv:2601.16376 (2026; preprint), arXiv · repository
PolyJarvis	Natural-language agent that connects an LLM to RadonPy through MCP to build, parameterize, equilibrate, and analyze all-atom polymer simulations. The authors report mixed accuracy across four polymers and eight comparisons.	Zhao, Chandrasekhar & Farimani, “PolyJarvis: LLM Agent for Autonomous Polymer MD Simulations,” arXiv:2604.02537 (2026; preprint), arXiv · repository

Resource	Scope	Source / status
ToPolyAgent	Natural-language multi-agent workflow for coarse-grained bead-spring LAMMPS simulations, including linear, ring, brush, star, and dendritic topologies.	Ding, Carrillo & Do, “ToPolyAgent: AI agents for coarse-grained bead-spring topological polymer simulations,” <i>Digital Discovery</i> 5 (2), 901–909 (2026), DOI · repository
MDCrow	General LLM-assisted MD workflow with 40 tools for simulation setup, file handling, analysis, and literature retrieval. The published benchmark focuses on protein simulations, not synthetic-polymer workflows.	Campbell, Cox, Medina, Watterson & White, “MDCrow: automating molecular dynamics workflows with large language models,” <i>Machine Learning: Science and Technology</i> 7 (2), 025037 (2026), DOI · 2025 preprint · code
G-BigSMILES 2.0	Stochastic, generative polymer representation that encodes distributions and complex architectures; can generate molecular ensembles and graph-learning inputs. This is a representation, not a simulation engine.	Tian, Zaldivar & Sun, “G-BigSMILES 2.0: A Generative Representation for Complex Polymer Architectures and Machine Learning,” APS Global Physics Summit 2026, oral presentation (16 Mar 2026), abstract

4.7 Notes on Scope

- A **force field** defines interactions; an **MD engine** integrates equations of motion; a **workflow package** automates some steps around an engine. These are related, but not interchangeable.
- UA and CG describe levels of molecular representation. TraPPE-UA, Kremer–Grest, and MARTINI are examples of specific parameterized models within those representations.
- When reporting software-based results, record the exact software release/commit, force-field family and parameter-set version, input structure, mapping, and validation protocol. A package or force-field family name alone is not enough to reproduce a calculation.
- For a new polymer chemistry, verify atom typing, charges, bonded parameters, non-bonded interactions, and any mapping choices. Report the validation data and simulation protocol with the results.
- Preprints, conference presentations, and peer-reviewed articles are labeled separately above. Recheck publication status when citing fast-moving 2026 work.

Chapter 5

Papers

Use this folder for literature notes, reading lists, citation collections, and short paper summaries related to polymer simulation.

5.1 VIP Paper

1. High-Throughput Molecular Dynamics Simulations and Validation of Thermophysical Properties of Polymers for Various Applications, ACS Appl. Polym. Mater. (2021) 3 (2): 620–630., <https://doi.org/10.1021/acsapm.0c00524>
2. Paudel, A., Shakya, A., Nag, S. et al. A structured dataset of polymers in molecular dynamics simulations extracted by LLM-assisted text mining. Sci Data (2026). <https://doi.org/10.1038/s41597-026-08045-x>
3. Macromolecules (2019) 52 (3): 755–786. <https://doi.org/10.1021/acs.macromol.8b01836>, Modeling and Simulations of Polymers: A Roadmap
4. Chew, A.K., Afzal, M.A.F., Kaplan, Z. et al. Leveraging high-throughput molecular simulations and machine learning for the design of chemical mixtures. npj Comput Mater 11, 72 (2025). <https://doi.org/10.1038/s41524-025-01552-2>
5. Atomistic polymer modeling: Recent advances and challenges in building and parametrization workflows HN Turney, M Matta - Macromolecules, 2025

5.2 Benchmark

1. Benchmarking Classical Molecular Dynamics Simulations for Computational Screening of Lithium Polymer Electrolytes, Macromolecules (2025) 58 (13): 6732–6742. <https://doi.org/10.1021/acs.macromol.5c00930>

5.3 High-Citation and Classic Reading List

Citation counts are approximate screening signals, because Google Scholar, ACS, Scopus, Crossref, OUCI, and other databases use different coverage. Counts below were checked from publicly visible pages on 2026-05-20.

Year	Paper	Why it matters	Citation signal
1985	Theodorou & Suter, <i>Detailed molecular structure of a vinyl polymer glass</i>	Early atomistic modeling of amorphous polymer glass structure; useful for packing, free volume, and glassy polymer model-building ideas.	ACS: cited by ~823 publications.
1990	Kremer & Grest, <i>Dynamics of entangled linear polymer melts: A molecular-dynamics simulation</i>	The foundational bead-spring / Kremer-Grest polymer melt model; essential for entanglement, Rouse-to-reptation crossover, and generic polymer MD.	OUCI: ~3,572 citations.
1990	Mayo, Olafson & Goddard, <i>DREIDING: A generic force field for molecular simulations</i>	General-purpose force field often encountered in polymer, organic, and materials simulations. Not polymer-specific, but important background for force-field choices.	ACS: cited by ~5,948 publications.
1997	Paul, Smith & Yoon, <i>Static and Dynamic Properties of a n-C100H202 Melt from Molecular Dynamics Simulations</i>	Representative atomistic/united-atom polyethylene melt simulation; good example of comparing MD chain statistics and dynamics with experiment.	Classic atomistic PE melt paper.
1998	Sun, <i>COMPASS: An Ab Initio Force-Field Optimized for Condensed-Phase Applications</i>	Widely used all-atom force field for condensed-phase simulations of organic molecules and polymers; common in Materials Studio/Forcite workflows.	CoLab: ~5,474 citations.

Year	Paper	Why it matters	Citation signal
1998	Bennemann, Paul, Binder & Dunweg, <i>Molecular-dynamics simulations of the thermal glass transition in polymer melts</i>	Classic dense polymer melt glass-transition MD study; useful for alpha relaxation, cooling protocols, and mode-coupling-theory comparisons.	Classic glass-transition simulation paper.
2000	Borodin & Smith, <i>Molecular Dynamics Simulations of Poly(ethylene oxide)/LiI Melts. 2. Dynamic Properties</i>	Important polymer-electrolyte MD paper; useful for ion transport, segmental dynamics, and PEO-salt coordination.	ACS: cited by ~115 publications.
2001	Akkermans & Briels, <i>A structure-based coarse-grained model for polymer melts</i>	A key bottom-up coarse-graining paper; highlights structure reproduction versus thermodynamic consistency.	University of Twente/Scopus: ~125 citations.
2002	Muller-Plathe, <i>Coarse-graining in polymer simulation: From the atomistic to the mesoscopic scale and back</i>	Review-style entry point for systematic polymer coarse graining and backmapping.	Classic coarse-graining review.
2003	Reith, Putz & Muller-Plathe, <i>Deriving effective mesoscale potentials from atomistic simulations</i>	Canonical iterative Boltzmann inversion style reference for deriving coarse-grained potentials from atomistic data.	Widely used coarse-graining method paper.
2003	Smith et al., <i>A molecular dynamics simulation study of nanoparticle interactions in a polymer-nanoparticle composite</i>	Early MD study of polymer-mediated nanoparticle interactions in a polymer matrix; useful for polymer nanocomposites.	ScienceDirect: cited by ~262 publications.

Year	Paper	Why it matters	Citation signal
2003	Starr, Douglas & Glotzer, <i>Origin of particle clustering in a simulated polymer nanocomposite and its impact on rheology</i>	Classic simulation paper on nanoparticle clustering, dispersion, shear, and rheology in polymer nanocomposites.	NIST-hosted publication; widely cited in PNC literature.
2004	Everaers et al., <i>Rheology and Microscopic Topology of Entangled Polymeric Liquids</i>	Primitive-path analysis foundation linking microscopic entanglement topology to rheology and plateau modulus.	Seminal Science paper.
2004	Faller, <i>Automatic coarse graining of polymers</i>	Clear review of automatic mapping and potential-optimization schemes for polymer coarse graining.	ScienceDirect: cited by ~102 publications.
2004	Paul & Smith, <i>Structure and dynamics of amorphous polymers: computer simulations compared to experiment and theory</i>	Broad review connecting atomistic and coarse-grained simulations with amorphous polymer experiments.	Broad review paper.
2006	Harmandaris et al., <i>Hierarchical Modeling of Polystyrene: From Atomistic to Coarse-Grained Simulations</i>	Practical multiscale workflow for polystyrene: atomistic model, coarse-graining, time mapping, and backmapping.	ACS: cited by ~334 publications.
2014	Liu et al., <i>Molecular dynamics simulation of dispersion and aggregation kinetics of nanorods in polymer nanocomposites</i>	Useful example for rod-like fillers, dispersion/aggregation kinetics, and CG polymer nanocomposite analysis.	ScienceDirect: cited by ~45 publications.
2017	Vogiatzis & Theodorou, <i>Multiscale Molecular Simulations of Polymer-Matrix Nanocomposites</i>	Open-access review of polymer-matrix nanocomposite simulations across length scales.	Springer: ~69 citations.

Year	Paper	Why it matters	Citation signal
2023	Shi, Qian & Lu, <i>Coarse-grained molecular dynamics simulation of polymers: Structures and dynamics</i>	Recent high-level review for CG polymer MD, including structure, dynamics, and chemically specific coarse-grained models.	Recent WIREs review.

5.4 Suggested Reading Path

1. Start with the generic melt baseline: Kremer & Grest 1990.
2. Add atomistic polymer examples: Theodorou & Suter 1985; Paul, Smith & Yoon 1997.
3. Learn force-field context: DREIDING 1990; COMPASS 1998.
4. Study glassy dynamics: Bennemann et al. 1998; Paul & Smith 2004.
5. Learn coarse graining: Akkermans & Briels 2001; Muller-Plathe 2002; Reith et al. 2003; Faller 2004; Harmandaris et al. 2006; Shi et al. 2023.
6. Study entanglement topology and rheology: Everaers et al. 2004.
7. Move to applications: Borodin & Smith 2000 for polymer electrolytes; Smith et al. 2003 and Starr et al. 2003 for polymer nanocomposites.

5.5 Tags to Use Later

- polymer-melt
- kremer-grest
- atomistic-md
- united-atom
- force-field
- coarse-graining
- backmapping
- glass-transition
- entanglement
- rheology
- polymer-electrolyte
- polymer-nanocomposite

5.6 To Expand

- Add BibTeX entries in papers/bib/.
- Add one Markdown note per paper in papers/notes/.
- Add a separate list for polymer crystallization, cross-linked polymers/epoxy, polymer membranes, and LAMMPS/RadonPy-specific workflows.